

Algoritem razvodja

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Implementacija algoritma razvodja

Algoritem razvodja je bil implementiran v **TypeScriptu** z uporabo **React** ogrodja za uporabniški vmesnik in prikaz rezultatov. Implementacija sledi principu simulacije poplav, kot je opisan v teoriji, pri čemer je vhod **predhodno obdelana sivinska slika**.

Algoritem poteka **po sivinskih nivojih (bucket by bucket)**, od najnižjih do najvišjih vrednosti. Za vsak sivinski nivo se obdelajo vsi piksli, ki pripadajo trenutnemu vedru. Je bila v tej implementaciji uporabljena **preiskava v širino (BFS)** znotraj posameznega vedra.

BFS se uporablja za sistematično obdelavo vseh pikslov v trenutnem sivinskem nivoju, pri čemer se širjenje regij začne iz že označenih sosednjih pikslov. To omogoča konsistentno in učinkovito širjenje regij znotraj istega vedra, hkrati pa ohranja logiko algoritma razvodja, kot je opisana v literaturi.

Dodeljevanje regij posameznemu pikslu poteka po naslednjih pravilih:

- če piksel nima nobenega sosedu z že določeno regijo, se mu dodeli nova regija,
- če ima natanko enega sosedu z določeno regijo, prevzame njegovo oznako,
- če ima dva ali več sosedov z različnimi regijami, se mu dodeli regija enega izmed sosedov.

Za zmanjšanje razdrobljenosti segmentacije je implementirano **odstranjevanje manjših robov**, pri čemer je prag za odstranjevanje uporabniško nastavljiv parameter. Končni rezultat algoritma je **segmentirana slika**, v kateri so posamezne regije barvno označene.

Rezultati – vpliv parametra za odstranjevanje robov

V nadaljevanju so prikazani rezultati algoritma razvodja za **tri različna območja**, pri čemer je za vsako območje prikazana segmentacija pri **treh različnih vrednostih parametra za odstranjevanje robov**.

Območje 1

Prag 1 (vrednost 10)

Watershed Algorithm - Edge Detection & Segmentation

Upload Image:

Edge Removal Threshold: 10

Choose file primer1.jpg

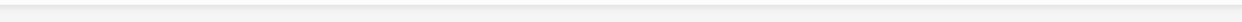
Original Image

Edge Detection (Morphological Gradient)

Cleaned Edges (After Threshold Removal)

Run Watershed Segmentation

Watershed Segmentation Result



Prag 2 (vrednost 20) => optimalna

Watershed Algorithm - Edge Detection & Segmentation

Upload Image:

Edge Removal Threshold: 20

Choose file

primer1.jpg

Original Image

Edge Detection (Morphological Gradient)

Cleaned Edges (After Threshold Removal)

Run Watershed Segmentation

Watershed Segmentation Result



Run Watershed Segmentation



Prag 3 (vrednost 30)

Watershed Algorithm - Edge Detection & Segmentation

Upload Image:

Edge Removal Threshold: 30

Choose file primer1.jpg

Original Image

Edge Detection (Morphological Gradient)

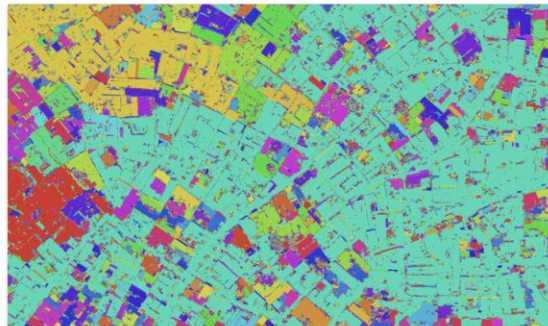
Cleaned Edges (After Threshold Removal)

Run Watershed Segmentation

Watershed Segmentation Result



Run Watershed Segmentation



Območje 2

Prag 1 (vrednost 10)

Watershed Algorithm - Edge Detection & Segmentation

Upload Image:

Edge Removal Threshold: 10

Choose file primer2.jpg

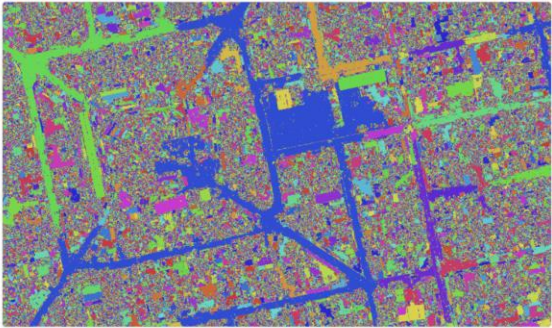
Original Image

Edge Detection (Morphological Gradient)

Cleaned Edges (After Threshold Removal)

Run Watershed Segmentation

Watershed Segmentation Result

The final result of the watershed segmentation algorithm. The image is divided into numerous small, irregular regions, each assigned a unique color. The colors are primarily shades of blue, green, and yellow, with some red and purple. The boundaries between these regions are clearly defined, following the edges of the original image.

Prag 2 (vrednost 17) => optimalna

Watershed Algorithm - Edge Detection & Segmentation

Upload Image:

Edge Removal Threshold: 17

Choose file

primer2.jpg

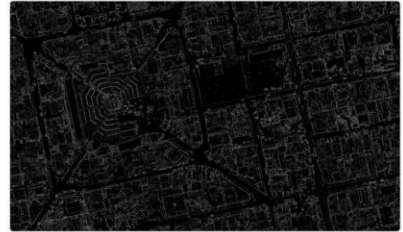
Original Image

Edge Detection (Morphological Gradient)

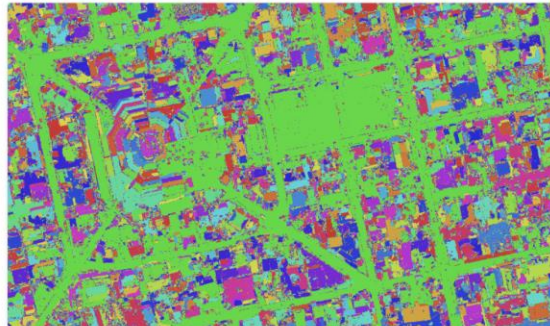
Cleaned Edges (After Threshold Removal)

Run Watershed Segmentation

Watershed Segmentation Result



Run Watershed Segmentation



Prag 3 (vrednost 29)

Watershed Algorithm - Edge Detection & Segmentation

Upload Image:

Edge Removal Threshold: 29

Choose file

primer2.jpg

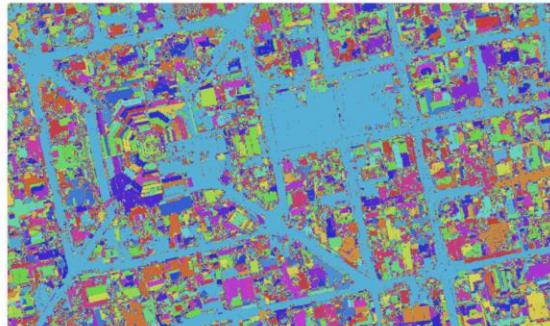
Original Image

Edge Detection (Morphological Gradient)

Cleaned Edges (After Threshold Removal)

Run Watershed Segmentation

Watershed Segmentation Result



Območje 3

Prag 1 (vrednost 10)

Watershed Algorithm - Edge Detection & Segmentation

Upload Image:

primer3.jpg

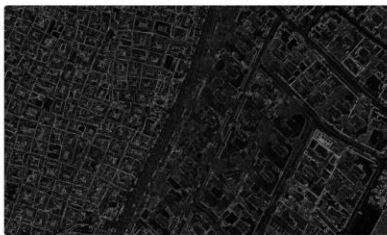
Edge Removal Threshold: 10



Original Image



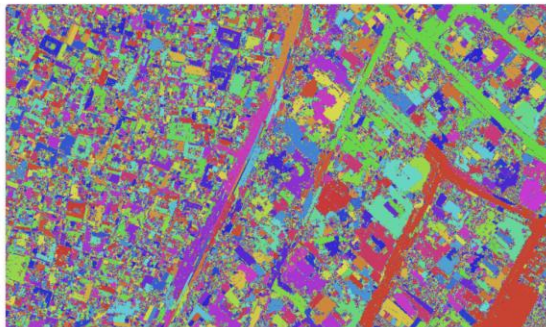
Edge Detection (Morphological Gradient)



Cleaned Edges (After Threshold Removal)



Watershed Segmentation Result



Prag 2 (vrednost 17) => optimalna

Watershed Algorithm - Edge Detection & Segmentation

Upload Image:

Edge Removal Threshold: 17

Choose file

primer3.jpg

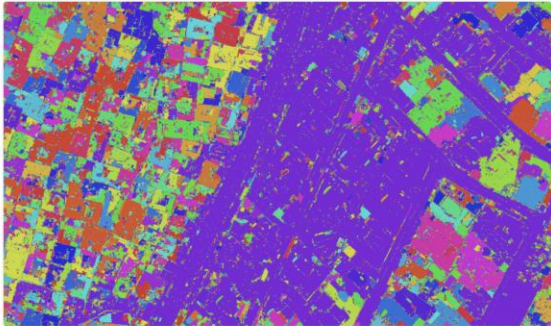
Original Image

Edge Detection (Morphological Gradient)

Cleaned Edges (After Threshold Removal)

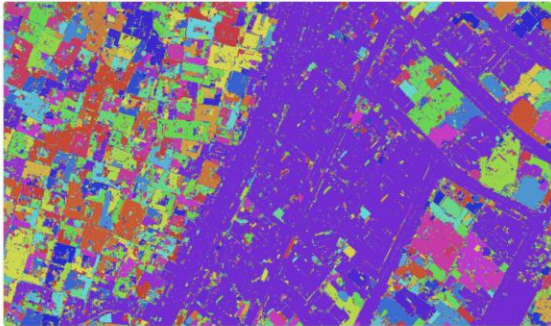
Run Watershed Segmentation

Watershed Segmentation Result





Run Watershed Segmentation



Prag 3 (vrednost 29)

Watershed Algorithm - Edge Detection & Segmentation

Upload Image:

Edge Removal Threshold: 29

Choose file

primer3.jpg

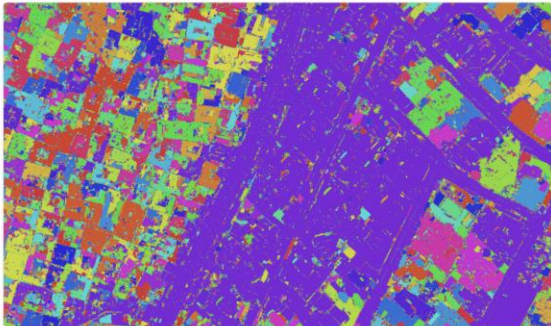
Original Image

Edge Detection (Morphological Gradient)

Cleaned Edges (After Threshold Removal)

Run Watershed Segmentation

Watershed Segmentation Result



The image displays a web-based interface for the Watershed Algorithm. It includes an upload section for an image (currently 'primer3.jpg') and a slider for the 'Edge Removal Threshold' (set to 29). Below these are three preview images: the 'Original Image' (aerial view of a city), 'Edge Detection (Morphological Gradient)' (a binary edge map), and 'Cleaned Edges (After Threshold Removal)' (a binary edge map with some noise removed). A blue button labeled 'Run Watershed Segmentation' is positioned below the edge maps. The final output is the 'Watershed Segmentation Result', which is a multi-colored image where different regions of the city are segmented into distinct colors (e.g., red, green, blue, yellow, purple).